

products and services. This action is slanted towards user belief mining or sentiment mining.

Key research dimensions in Web scraping

- **Cyber patrol:** This entails analysis of messages on social media about particular crimes, or of discussions about particular crimes.
- **Cyber parenting:** This is the extraction of data from Web portals that engage children or teenagers, so that their emotions and behaviour can be analysed. The cyber parenting approaches that guard children against inappropriate portals can be integrated with these implementations.
- **Corporate applications:** This involves fetching reviews of products and services from e-commerce portals so that the companies can improve their products as per the requirements and feedback of the users.

Prominent tools for Web scraping and real-time data extraction

There are a number of tools and libraries available for the extraction of real-time data from Web pages. Table 1 mentions the assorted tools, along with the associated URL from where these libraries can be installed.

Python based tools for Web scraping

For the last few years, Python has gained much prominence as a high performance programming language for multiple applications. Python is nowadays used in cyber security, digital forensics, Web analytics, deep learning, machine learning, grid computing, parallel computing, cloud applications, Web scraping and many other areas in which a higher degree of performance and accuracy are needed, with minimal error rates.

Python has more than 200,000 packages in its repository, *pypi.org*,

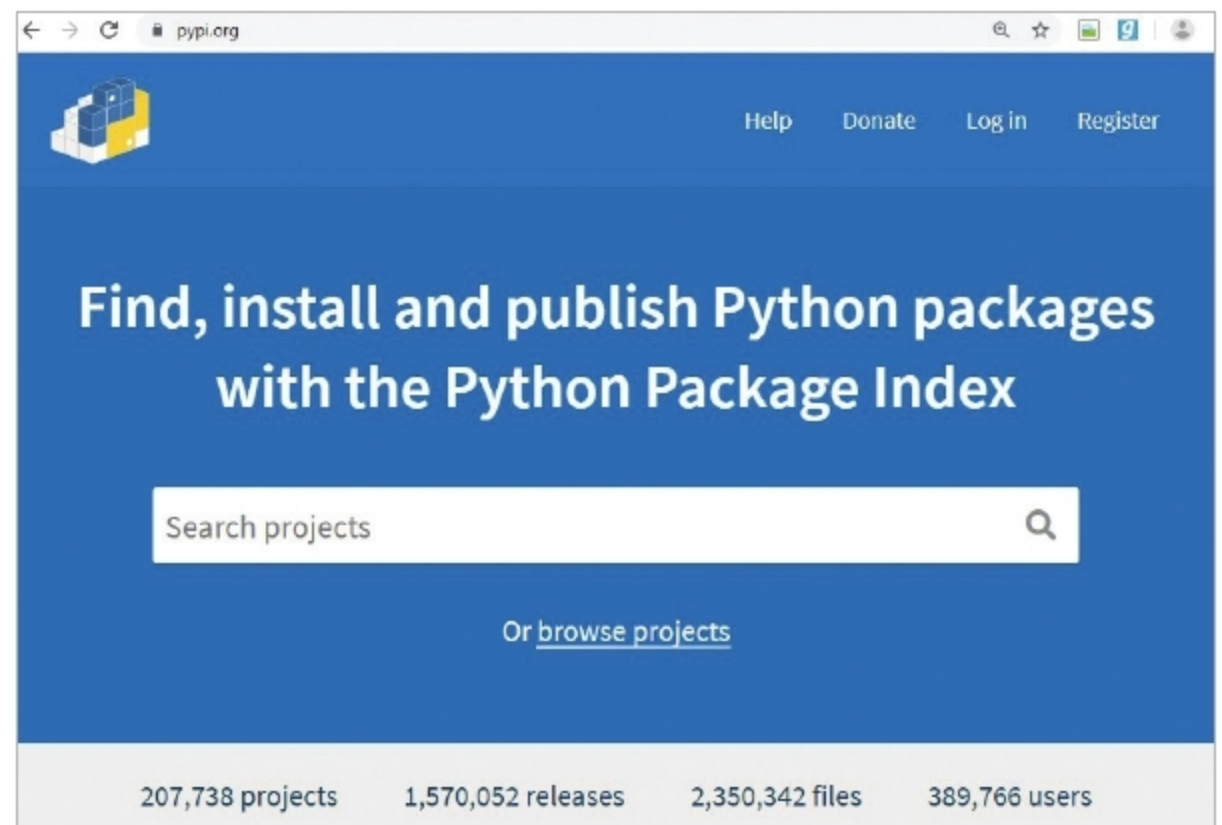


Figure 1: *pypi.org* as the key repository of Python packages

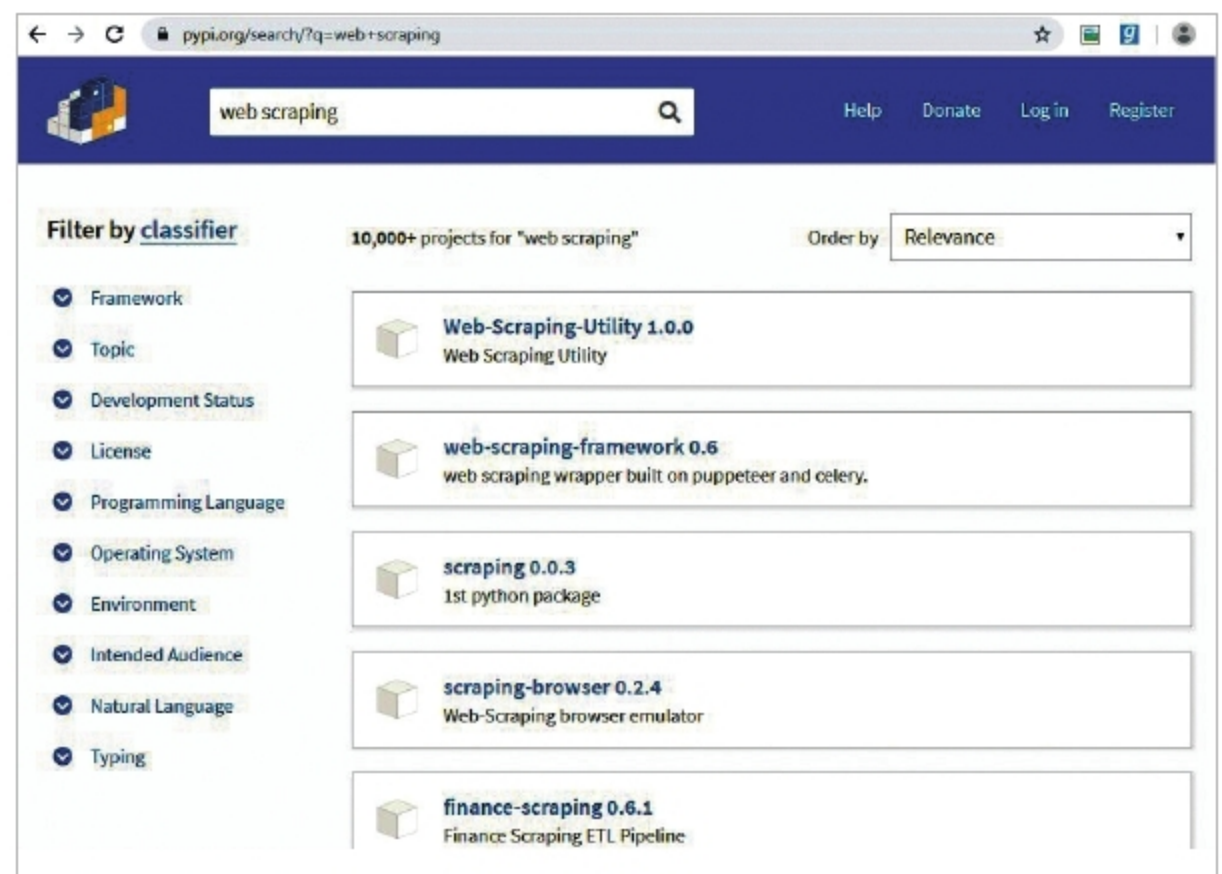


Figure 2: Packages for Web scraping on *pypi.org*

which enable programmers and researchers to work on diversified domains with immense flexibility.

For Web scraping and real-time data extraction, there are more than 10,000 projects and code libraries that can be used by developers.

The difference between Web parsers and Web scrapers

Table 2 lists popular Web parsers

for Web applications, which are used to analyse the scraped data. Essentially, a Web scraper is used to extract data from Web applications, while the Web parser is used to break down and analyse the scraped data. So after Web scrapers are used, Web parsers are integrated to analyse the extracted data for knowledge discovery and storage for particular applications.